

POWERFACTORY

PowerFactory 2024

Technical Reference

PLL Phase/Frequency Measurement

StaPll

PF2024

POWER SYSTEM SOLUTIONS

MADE IN GERMANY

Publisher:
DIgSILENT GmbH
Heinrich-Hertz-Straße 9
72810 Gomaringen / Germany
Tel.: +49 (0) 7072-9168-0
Fax: +49 (0) 7072-9168-88
info@digsilent.de

Please visit our homepage at:
https://www.digsilent.de

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1 General Description

Phase-locked loops (PLLs) are widely used for synchronization purposes, controller applications and can also be used to demodulate frequency-modulated signals. For more information about PLL definition and applications refer to [1] and [2].

The *PLL Phase/Frequency Measurement* element (*StaPll*) measures the frequency and phase of a selected voltage in the system.

The PLL model is supported by the RMS and EMT simulations only.

In the following sections the working and behaviour of the PLL is presented.

2 RMS-Simulation

Depending on the selected *No. of phases* option (parameter $nphase$), a three-phase or single-phase measurement is possible.

The PLL uses the voltage at the selected *Measurement point* (reference node) as inputs in the input dialogue box of the PLL.

2.1 Three-phase measurement

The PLL structure is the same for both balanced and unbalanced RMS simulations. Internally, the input signals to the PLL are the real and imaginary part of the positive sequence voltage (in-quadrature signals). In the case of balanced network representation, the positive sequence voltage resulting from the simulation is fed directly to the input signals (uR, uI) of the PLL. In the case of unbalanced network representation, the positive sequence voltage is first being calculated using the symmetrical transformation on the input signals fed to the PLL $(uR_A, uI_A, uR_B, uI_B, uR_C, uI_C)$. The input signals including the frequency of the reference machine in per unit $fref$ are connected automatically.

The block diagram of the PLL is presented in Figure 2.1.

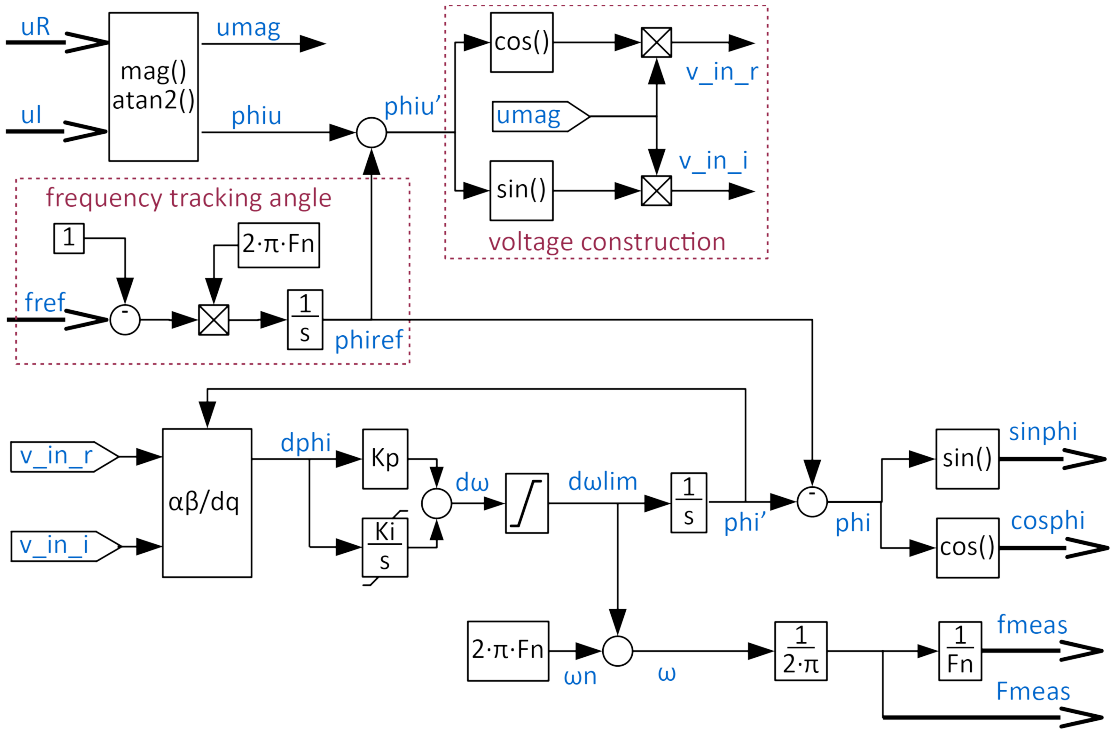


Figure 2.1: RMS simulation model of the SRF PLL

The measured in-quadrature voltage input signals uR and uI are in the coordinate system of the reference angle. The influence of the reference angle (and reference frequency) is eliminated in order to obtain the local measured voltage without this external influence. This is done by using a tracking angle $phiref$ (between the measured frequency input $fref$ and the nominal frequency), to construct the voltage inputs for the $\alpha\beta/dq$ transformation.

The input to the PI controller $(dphi)$ is the q-component of the dq-voltage obtained from the

constructed voltage as:

$$vd = v_in_r \cdot cos(phi') + v_in_i \cdot sin(phi') \quad (1)$$

$$vq = v_in_i \cdot cos(phi') - v_in_r \cdot sin(phi') \quad (2)$$

The PI controller is implemented according to IEEE standard 421.5-2016. Depending if the output of the PI controller $d\omega$ is in limit or not, the following is valid for the output of the limiter $d\omega lim$ and for the state variable xpi of the PI controller:

$$d\omega lim = \begin{cases} d\omega max & \text{upper limit if } d\omega > d\omega max \text{ and } dphi > 0 \\ Kp \cdot dphi + xpi & \text{normal operation} \\ d\omega min & \text{lower limit if } d\omega < d\omega min \text{ and } dphi < 0 \end{cases} \quad (3)$$

$$\frac{d}{dt}xpi = \begin{cases} 0 & \text{upper limit if } d\omega > d\omega max \text{ and } dphi > 0 \\ Ki \cdot dphi & \text{normal operation} \\ 0 & \text{lower limit if } d\omega < d\omega min \text{ and } dphi < 0 \end{cases} \quad (4)$$

where the limits in rad/s are $d\omega max = (fmax - 1) \cdot 2 \cdot \pi \cdot Fn$ and $d\omega min = (fmin - 1) \cdot 2 \cdot \pi \cdot Fn$ and:

- $fmin$ is the *Lower frequency limit* parameter in p.u.;
- $fmax$ is the *Upper frequency limit* parameter in p.u.;
- Fn is the nominal frequency in Hz;
- Kp is the proportional gain of the PI controller;
- Ki is the integral gain of the PI controller;

2.1.1 Input voltage normalisation

If the *Input voltage normalisation* is enabled, the input to the PI controller $dphi$ (vq voltage) is divided by the magnitude of the dq voltage $vdq = \sqrt{vd^2 + vq^2}$.

2.1.2 Blocking functionality

In real devices, the angle cannot be tracked if the voltage falls below a certain threshold. The option *Voltage dependent blocking* can be used to simulate this. If the option is enabled, the *Blocking voltage* input parameter $umin$ is compared against the voltage magnitude calculated as:

$$umag = \sqrt{uR^2 + uI^2} \quad (5)$$

If the voltage magnitude gets below the blocking voltage, the blocking state is enabled and the input to the PI controller is set to zero $dphi = 0$.

If in addition the option *Set the angular frequency deviation to zero when blocked* is enabled, the angular frequency deviation is set to zero $dwlim = 0$.

2.1.3 Signals

The signals used in the case of a balanced network representation are presented in Table 2.1. The input signals are automatically connected (internally in *PowerFactory*).

Table 2.1: Signals for three-phase measurement (balanced RMS-Simulation)

Name	Symbol	Unit	Type	Description
$Fmeas$		Hz	OUT	Measured Frequency
$fmeas$		p.u.	OUT	Measured Frequency
$cosphi$			OUT	$\cos(phi)$
$sinphi$			OUT	$\sin(phi)$
xpi			STATE	State variable of the PI controller
$xphi$		rad	STATE	State variable of the integrator
uR		p.u.	IN	Input Voltage, Real Part
uI		p.u.	IN	Input Voltage, Imaginary Part
$fref$		p.u.	IN	Reference Frequency

The signals used in the case of an unbalanced network representation are presented in Table 2.2. The input signals are automatically connected (internally in *PowerFactory*).

Table 2.2: Signals for three-phase measurement (unbalanced RMS-Simulation)

Name	Symbol	Unit	Type	Description
$Fmeas$		Hz	OUT	Measured Frequency
$fmeas$		p.u.	OUT	Measured Frequency
$cosphi$			OUT	$\cos(phi)$
$sinphi$			OUT	$\sin(phi)$
xpi			STATE	State variable of the PI controller
$xphi$		rad	STATE	State variable of the integrator (Phi)
uR		p.u.	IN	Input Voltage, Real Part
uI		p.u.	IN	Input Voltage, Imaginary Part
uR_A		p.u.	IN	Input Voltage, Phase a, Real Part
uI_A		Hz	IN	Input Voltage, Phase a, Imaginary Part
uR_B		p.u.	IN	Input Voltage, Phase b, Real Part
uI_B		Hz	IN	Input Voltage, Phase b, Imaginary Part
uR_C		p.u.	IN	Input Voltage, Phase c, Real Part
uI_C		Hz	IN	Input Voltage, Phase c, Imaginary Part
$fref$		p.u.	IN	Reference Frequency

2.2 Single-phase measurement

Depending on the *Phase technology* of the selected *Measurement point*, the single-phase PLL can measure at phase a, b, c, $a-n$, $b-n$ or $c-n$. This can be selected by using the *Measured phase* parameter.

The in-quadrature signals fed to the PLL algorithm are then automatically selected from the phase voltages depending on the *Measured phase*.

The block diagram of the single-phase PLL is equal to the three-phase PLL diagram (presented in Figure 2.1).

2.2.1 Input voltage normalisation

The normalisation of the input voltage is equivalent as for the three-phase measurement. Please refer to Section 2.1.1.

2.2.2 Blocking functionality

The blocking functionality is equivalent as for the three-phase measurement. Please refer to Section 2.1.2.

2.2.3 Signals

The signals used in the single-phase RMS model are the same as the signals used for the three-phase measurement in the balanced RMS simulation presented in Table 2.1.

3 EMT-Simulation

Depending on the selected *No. of phases* option, a three-phase or single-phase PLL measurement is possible.

3.1 Three-phase measurement

There are two different PLL types available in the *StaPll* for the EMT simulation:

- SRF PLL,
- DSOGI PLL.

3.1.1 SRF PLL

The block diagram of the synchronous reference frame based PLL is presented in Figure 3.1.

The phase voltages available from the EMT-simulation are used by the $abc/\alpha\beta$ transformation to obtain the $\alpha\beta$ voltage components:

$$valpha_in = \frac{2}{3} \cdot u_a - \frac{1}{3} \cdot u_b - \frac{1}{3} \cdot u_c \tag{6}$$

$$vbeta_in = 0 \cdot u_a + \frac{\sqrt{3}}{3} \cdot u_b - \frac{\sqrt{3}}{3} \cdot u_c \tag{7}$$

The input to the PI controller ($dphi$) is the q-component of the dq-voltage calculated from $ualpha_in$ and $ubeta_in$:

$$vd = valpha_in \cdot cos(phi) + vbeta_in \cdot sin(phi) \tag{8}$$

$$vq = vbeta_in \cdot cos(phi) - valpha_in \cdot sin(phi) \tag{9}$$

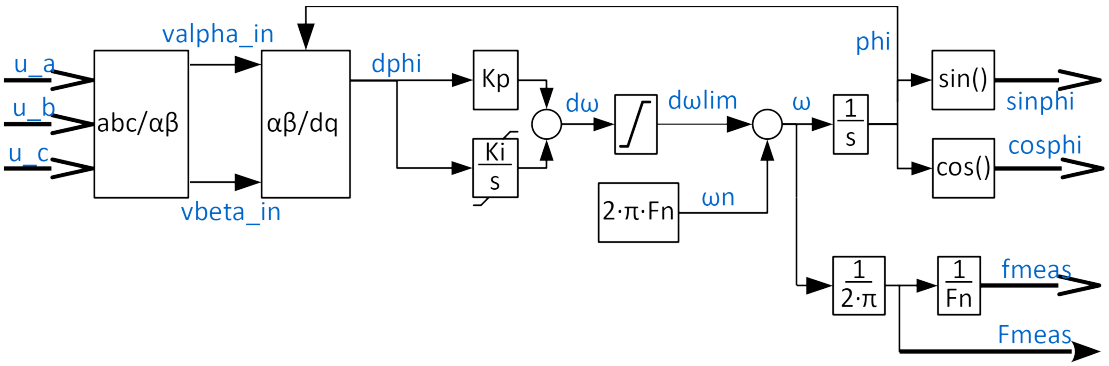


Figure 3.1: EMT simulation model of the SRF PLL

The PI controller is implemented according to IEEE standard 421.5-2016. Depending if the output of the PI controller $d\omega$ is in limit or not, the following is valid for the output of the limiter

$d\omega lim$ and for the derivative of the state variable xpi of the PI controller:

$$d\omega lim = \begin{cases} d\omega max & \text{upper limit if } d\omega > d\omega max \text{ and } dphi > 0 \\ Kp \cdot dphi + xpi & \text{normal operation} \\ d\omega min & \text{lower limit if } d\omega < d\omega min \text{ and } dphi < 0 \end{cases} \quad (10)$$

$$\frac{d}{dt}xpi = \begin{cases} 0 & \text{upper limit if } d\omega > d\omega max \text{ and } dphi > 0 \\ Ki \cdot dphi & \text{normal operation} \\ 0 & \text{lower limit if } d\omega < d\omega min \text{ and } dphi < 0 \end{cases} \quad (11)$$

where the limits in rad/s are $d\omega max = (fmax - 1) \cdot 2 \cdot \pi \cdot Fn$ and $d\omega min = (fmin - 1) \cdot 2 \cdot \pi \cdot Fn$ and:

- $fmin$ is the *Lower frequency limit* parameter in p.u.;
- $fmax$ is the *Upper frequency limit* parameter in p.u.;
- Fn is the nominal frequency in Hz;
- Kp is the proportional gain of the PI controller;
- Ki is the integral gain of the PI controller;

3.1.1.1 Notch filter for oscilations at double nominal angular frequency

The notch filter is filtering the the dq-voltages vd and vq that are obtained from the $\alpha\beta/dq$ transformation (Figure 3.2).

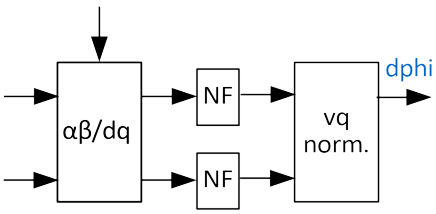


Figure 3.2: Notch filter placement in the block diagram of the PLL

The following transfer function is used for the notch filter:

$$\frac{yo(s)}{yi(s)} = \frac{s^2 + wnf^2}{s^2 + s \cdot bw + wnf^2} \quad (12)$$

where :

- wnf is the notch filter angular frequency (rejected frequency) in rad/s which is always set to double nominal angular frequency $2 \cdot \omega n = 2 \cdot 2 \cdot \pi \cdot Fn$;
- bw is the bandwidth of the notch filter (width of rejected band) input parameter in rad/s. The default value is set to 10% of the 50Hz wnf value;

3.1.1.2 Input voltage normalisation

If the *Input voltage normalisation* is enabled, the input to the PI controller $dphi$ (vq voltage) is divided by the magnitude of the dq voltage $vdq = \sqrt{vd^2 + vq^2}$ (obtained from the $\alpha\beta/dq$ transformation or from the notch filter if it is enabled).

3.1.1.3 Blocking functionality

In real devices, the angle cannot be tracked if the voltage falls below a certain threshold. The option *Voltage dependent blocking* can be used to simulate this. If the option is enabled, the *Blocking voltage* input parameter $umin$ is compared against the voltage magnitude calculated as:

$$umag = \sqrt{valpha_in^2 + vbeta_in^2} \tag{13}$$

If the voltage magnitude gets below the blocking voltage, the blocking state is enabled and the input to the PI controller is set to zero ($dphi = 0$).

3.1.2 DSOGI PLL

The double second-order generalized integrator (DSOGI) based PLL gives more accurate phase and frequency in the case of unbalanced and harmonic grid voltage conditions.

The input to the PI controller $dphi$ (the q-component of the dq-voltage) is calculated using $ualpha_pos$ and $ubeta_pos$. The rest of the logic is same as for the SRF PLL.

The block diagram showing the calculation of the $\alpha\beta$ voltage-components of the SOGI PLL is presented in Figure 3.3.

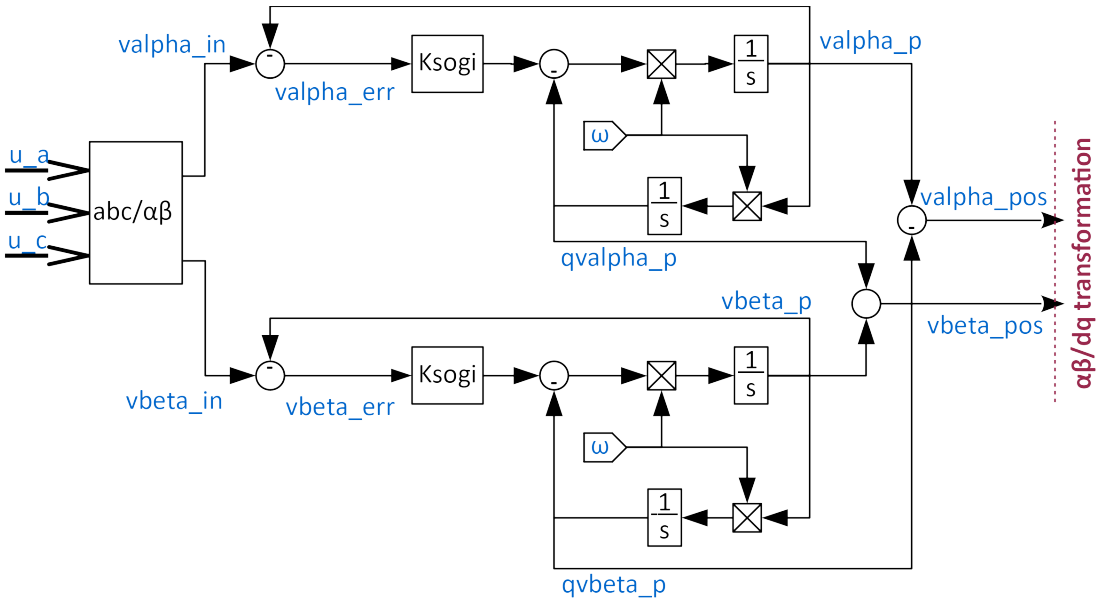


Figure 3.3: EMT simulation model of the DSOGI PLL

3.1.2.1 Input voltage normalisation

The input voltage normalisation is identical as in Section 3.1.1.2.

3.1.2.2 Blocking functionality

The blocking functionality is identical as in Section 3.1.1.3.

3.1.3 Signals

The signals used for the three-phase measurement in the EMT model are presented in Table 3.1. The input signals are automatically connected (internally in *PowerFactory*).

Table 3.1: Signals for three-phase measurement (EMT-Simulation)

Name	Symbol	Unit	Type	Description
$Fmeas$		Hz	OUT	Measured Frequency
$fmeas$		p.u.	OUT	Measured Frequency
$cosphi$			OUT	$\cos(phi)$
$sinphi$			OUT	$\sin(phi)$
xpi			STATE	State variable of the PI controller
$xphi$		rad	STATE	State variable of the integrator (phi)
u_a		p.u.	IN	Input Voltage, Phase a
u_b		p.u.	IN	Input Voltage, Phase b
u_c		p.u.	IN	Input Voltage, Phase c

3.2 Single-phase measurement

The single-phase PLL can measure at phase a, b, c, $a-n$, $b-n$ or $c-n$.

There are two different PLL types available in the *StaPll*:

- Inverse Park PLL,
- SOGI PLL.

3.2.1 Inverse Park PLL

Since the EMT simulation provides only one voltage and the PLL algorithm requires in-quadrature signals, the second signal has to be constructed. The in-quadrature image of the input single-phase signal is created using the Inverse Park Transform [2]. The image signal is constructed by introducing two low-pass filters in a loop consisting of the direct and inverse Park transformations. The filters use the *Low-pass filter time constant* $T1$ in ms. The default value of $T1$ is selected for a $50Hz$ system [2]. The rest of the logic is same as for the SRF PLL.

The block diagram showing the inverse Park based PLL is presented in Figure 3.4.

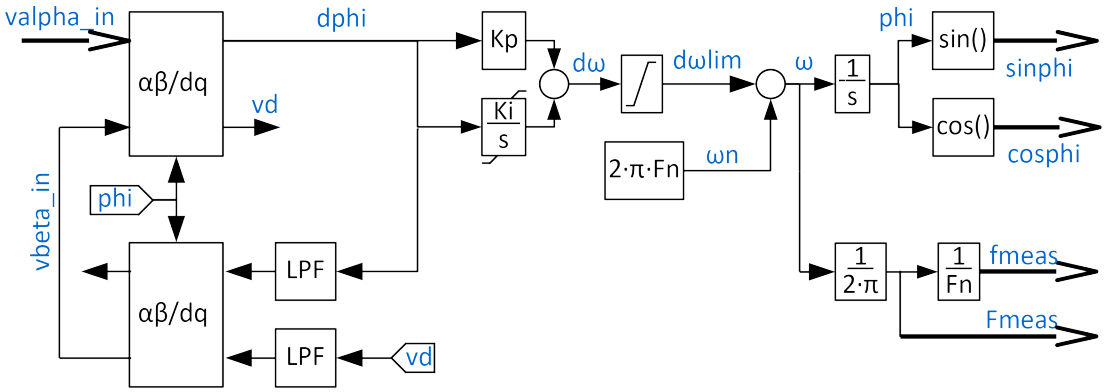


Figure 3.4: EMT simulation model of the inverse Park PLL

The variable $valpha_in = uR$, or if the neutral connection is measured $valpha_in = uR - uN$.

3.2.1.1 Input voltage normalisation

The input voltage normalisation is identical as in Section 3.1.1.2.

3.2.1.2 Blocking functionality

The blocking functionality is identical as in Section 3.1.1.3.

3.2.2 SOGI PLL

Similar ss in the case of the inverse park based PLL, the in-quadrature signal of the input single-phase signal is created by the second-order generalized integrator (SOGI). The input to the PI controller $dphi$ (the q-component of the dq-voltage) is calculated using $valpha$ and $qvalpha$. The rest of the logic is same as for the SRF PLL.

The block diagram showing the calculation of the $\alpha\beta$ voltage-components for the SOGI PLL is presented in Figure 3.5.

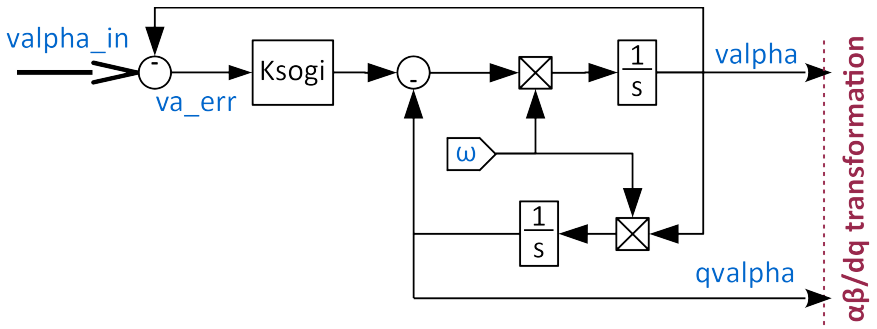


Figure 3.5: EMT simulation model of the SOGI PLL

The variable $valpha_in = uR$, or if the neutral connection is measured $valpha_in = uR - uN$.

3.2.2.1 Input voltage normalisation

The input voltage normalisation is identical as in Section 3.1.1.2.

3.2.2.2 Blocking functionality

The blocking functionality is identical as in Section 3.1.1.3.

3.2.2.3 Input voltage normalisation

The input voltage normalisation is identical as in Section 3.1.1.2.

3.2.2.4 Blocking functionality

.

The blocking functionality is identical as in Section 3.1.1.3.

3.2.3 Signals

The signals used for the single-phase measurement in the EMT model are presented in Table 3.2. The input signal is automatically connected (internally in *PowerFactory*).

Table 3.2: Signals for single-phase Measurement (EMT-Simulation)

Name	Symbol	Unit	Type	Description
$Fmeas$		Hz	OUT	Measured Frequency
$fmeas$		p.u.	OUT	Measured Frequency
$cosphi$			OUT	$\cos(phi)$
$sinphi$			OUT	$\sin(phi)$
xpi			STATE	State variable of the PI controller
$xphi$		rad	STATE	State variable of the integrator (Phi)
uR		p.u.	IN	Input Voltage

4 References

[1] R. E. Best. *Phase-Locked Loops: Design, Simulation and Applications*. McGraw Hill, 6 edition, 2007.

[2] R. Teodorescu, M. Liserre, and P. Rodriguez. *Grid Converters for Photovoltaic and Wind Power Systems*. Wiley, 2011.

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